

OOP Practical - 1

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Topic- Create a C++ program for Pascal's triangle using the mathematical concepts.

Date - 03/08/2026

Code :-

```
#include <iostream>
using namespace std;

// Function to calculate factorial
int factorial(int n)
{
    int fact = 1;
    for(int i = 1; i <= n; i++)
    {
        fact = fact * i;
    }
    return fact;
}

int main()
{
    int rows;

    cout << "Enter number of rows: ";
    cin >> rows;

    for(int n = 0; n < rows; n++)
    {
        for(int space = 0; space < rows - n - 1; space++)
        {
            cout << " ";
        }

        for(int r = 0; r <= n; r++)
        {
            cout << factorial(n) / (factorial(r) * factorial(n - r)) << " ";
        }
    }
}
```

```
        cout << endl;  
    }  
  
    return 0;  
}
```

Output:-

```
Enter number of rows: 7  
    1  
  1 1  
1 2 1  
1 3 3 1  
1 4 6 4 1  
1 5 10 10 5 1  
1 6 15 20 15 6 1  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```